

Data Profiling

load the required library

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import zscore
import seaborn as sns
```

define the path and read the dataset

```
In [2]: data_path_sierraleone = "C:\\Users\\girma\\OneDrive\\Desktop\\10 Academy\\week-1\\s
data_path_benin = "C:\\Users\\girma\\OneDrive\\Desktop\\10 Academy\\week-1\\s
data_path_togo = "C:\\Users\\girma\\OneDrive\\Desktop\\10 Academy\\week-1\\s

benin = pd.read_csv(data_path_benin )
sierraleone = pd.read_csv(data_path_sierraleone)
togo = pd.read_csv(data_path_togo)
```

Makesure the Timestamp in the correct datetime format

```
In [5]: benin['Timestamp'] = pd.to_datetime(benin['Timestamp'])
sierraleone['Timestamp'] = pd.to_datetime(sierraleone['Timestamp'])
togo['Timestamp'] = pd.to_datetime(togo['Timestamp'])
```

View first few rows

Sierraleone solar dataset

```
In [12]: sierraleone.head()
```

Out[12]:

	Timestamp	GHI	DNI	DHI	ModA	ModB	Tamb	RH	WS	WSgust	WSstdev	WD
0	2021-10-30 00:01:00	-0.7	-0.1	-0.8	0.0	0.0	21.9	99.1	0.0	0.0	0.0	0.0
1	2021-10-30 00:02:00	-0.7	-0.1	-0.8	0.0	0.0	21.9	99.2	0.0	0.0	0.0	0.0
2	2021-10-30 00:03:00	-0.7	-0.1	-0.8	0.0	0.0	21.9	99.2	0.0	0.0	0.0	0.0
3	2021-10-30 00:04:00	-0.7	0.0	-0.8	0.0	0.0	21.9	99.3	0.0	0.0	0.0	0.0
4	2021-10-30 00:05:00	-0.7	-0.1	-0.8	0.0	0.0	21.9	99.3	0.0	0.0	0.0	0.0

Benin solar dataset

In [9]:

benin.head()

Out[9]:

	Timestamp	GHI	DNI	DHI	ModA	ModB	Tamb	RH	WS	WSgust	WSstdev	WD
0	2021-08-09 00:01:00	-1.2	-0.2	-1.1	0.0	0.0	26.2	93.4	0.0	0.4	0.1	122.1
1	2021-08-09 00:02:00	-1.1	-0.2	-1.1	0.0	0.0	26.2	93.6	0.0	0.0	0.0	0.0
2	2021-08-09 00:03:00	-1.1	-0.2	-1.1	0.0	0.0	26.2	93.7	0.3	1.1	0.5	124.6
3	2021-08-09 00:04:00	-1.1	-0.1	-1.0	0.0	0.0	26.2	93.3	0.2	0.7	0.4	120.3
4	2021-08-09 00:05:00	-1.0	-0.1	-1.0	0.0	0.0	26.2	93.3	0.1	0.7	0.3	113.2

Togo solar dataset

In [11]:

togo.head()

Out[11]:

	Timestamp	GHI	DNI	DHI	ModA	ModB	Tamb	RH	WS	WSgust	WSstddev	WD
0	2021-10-25 00:01:00	-1.3	0.0	0.0	0.0	0.0	24.8	94.5	0.9	1.1	0.4	227.6
1	2021-10-25 00:02:00	-1.3	0.0	0.0	0.0	0.0	24.8	94.4	1.1	1.6	0.4	229.3
2	2021-10-25 00:03:00	-1.3	0.0	0.0	0.0	0.0	24.8	94.4	1.2	1.4	0.3	228.5
3	2021-10-25 00:04:00	-1.2	0.0	0.0	0.0	0.0	24.8	94.3	1.2	1.6	0.3	229.1
4	2021-10-25 00:05:00	-1.2	0.0	0.0	0.0	0.0	24.8	94.0	1.3	1.6	0.4	227.5

Check data types and values of the datatypes

In [16]:

```
sierraleone.info(); benin.info();togo.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525600 entries, 0 to 525599
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Timestamp              525600 non-null  datetime64[ns]
1   GHI                    525600 non-null  float64
2   DNI                    525600 non-null  float64
3   DHI                    525600 non-null  float64
4   ModA                   525600 non-null  float64
5   ModB                   525600 non-null  float64
6   Tamb                   525600 non-null  float64
7   RH                     525600 non-null  float64
8   WS                     525600 non-null  float64
9   WSGust                 525600 non-null  float64
10  WSstdev                525600 non-null  float64
11  WD                     525600 non-null  float64
12  WDstdev                525600 non-null  float64
13  BP                     525600 non-null  int64
14  Cleaning               525600 non-null  int64
15  Precipitation          525600 non-null  float64
16  TModA                  525600 non-null  float64
17  TModB                  525600 non-null  float64
18  Comments                0 non-null       float64
dtypes: datetime64[ns](1), float64(16), int64(2)
memory usage: 76.2 MB
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525600 entries, 0 to 525599
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Timestamp              525600 non-null  datetime64[ns]
1   GHI                    525600 non-null  float64
2   DNI                    525600 non-null  float64
3   DHI                    525600 non-null  float64
4   ModA                   525600 non-null  float64
5   ModB                   525600 non-null  float64
6   Tamb                   525600 non-null  float64
7   RH                     525600 non-null  float64
8   WS                     525600 non-null  float64
9   WSGust                 525600 non-null  float64
10  WSstdev                525600 non-null  float64
11  WD                     525600 non-null  float64
12  WDstdev                525600 non-null  float64
13  BP                     525600 non-null  int64
14  Cleaning               525600 non-null  int64
15  Precipitation          525600 non-null  float64
16  TModA                  525600 non-null  float64
17  TModB                  525600 non-null  float64
18  Comments                0 non-null       float64
dtypes: datetime64[ns](1), float64(16), int64(2)
memory usage: 76.2 MB
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 525600 entries, 0 to 525599
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
```

```
-----
0  Timestamp      525600 non-null  datetime64[ns]
1  GHI            525600 non-null  float64
2  DNI            525600 non-null  float64
3  DHI            525600 non-null  float64
4  ModA           525600 non-null  float64
5  ModB           525600 non-null  float64
6  Tamb           525600 non-null  float64
7  RH             525600 non-null  float64
8  WS             525600 non-null  float64
9  WSgust         525600 non-null  float64
10 WSstdev        525600 non-null  float64
11 WD             525600 non-null  float64
12 WDstdev        525600 non-null  float64
13 BP             525600 non-null  int64
14 Cleaning       525600 non-null  int64
15 Precipitation  525600 non-null  float64
16 TModA          525600 non-null  float64
17 TModB          525600 non-null  float64
18 Comments       0 non-null      float64
dtypes: datetime64[ns](1), float64(16), int64(2)
memory usage: 76.2 MB
```

Summary Statistics & Missing-Value Analysis

summary statistics for numeric columns

Summary statistic for Sierraleone Solar Dataset

```
In [18]: sierraleone.describe().T
```

Out[18]:

	count	mean	min	25%	50%	75%	max
Timestamp	525600	2022-04-30 12:00:30.000000768	2021-10-30 00:01:00	2022-01-29 06:00:45	2022-04-30 12:00:30	2022-07-30 18:00:15	2022-10-30 00:00:00
GHI	525600.0	201.957515	-19.5	-2.8	0.3	362.4	1499.0
DNI	525600.0	116.376337	-7.8	-0.3	-0.1	107.0	946.0
DHI	525600.0	113.720571	-17.9	-3.8	-0.1	224.7	892.0
ModA	525600.0	206.643095	0.0	0.0	3.6	359.5	1507.0
ModB	525600.0	198.114691	0.0	0.0	3.4	345.4	1473.0
Tamb	525600.0	26.319394	12.3	23.1	25.3	29.4	39.9
RH	525600.0	79.448857	9.9	68.7	85.4	96.7	100.0
WS	525600.0	1.146113	0.0	0.0	0.8	2.0	19.2
WSgust	525600.0	1.691606	0.0	0.0	1.6	2.6	23.9
WSstdev	525600.0	0.363823	0.0	0.0	0.4	0.6	4.1
WD	525600.0	133.044668	0.0	0.0	161.5	234.1	360.0
WDstdev	525600.0	7.17222	0.0	0.0	6.2	12.0	98.4
BP	525600.0	999.876469	993.0	999.0	1000.0	1001.0	1006.0
Cleaning	525600.0	0.000967	0.0	0.0	0.0	0.0	1.0
Precipitation	525600.0	0.004806	0.0	0.0	0.0	0.0	2.4
TModA	525600.0	32.504263	10.7	23.5	26.6	40.9	72.8
TModB	525600.0	32.593091	11.1	23.8	26.9	41.3	70.4
Comments	0.0	NaN	NaN	NaN	NaN	NaN	NaN

Summary statistic for Benin Solar Dataset

In [19]: `benin.describe().T`

Out[19]:

	count	mean	min	25%	50%	75%	max	
Timestamp	525600	2022-02-07 12:00:30.000000512	2021-08-09 00:01:00	2021-11-08 06:00:45	2022-02-07 12:00:30	2022-05-09 18:00:15	2022-08-09 00:00:00	
GHI	525600.0	240.559452	-12.9	-2.0	1.8	483.4	1413.0	3
DNI	525600.0	167.187516	-7.8	-0.5	-0.1	314.2	952.3	2
DHI	525600.0	115.358961	-12.6	-2.1	1.6	216.3	759.2	1
ModA	525600.0	236.589496	0.0	0.0	4.5	463.7	1342.3	3
ModB	525600.0	228.883576	0.0	0.0	4.3	447.9	1342.3	3
Tamb	525600.0	28.179683	11.0	24.2	28.0	32.3	43.8	
RH	525600.0	54.487969	2.1	28.8	55.1	80.1	100.0	
WS	525600.0	2.121113	0.0	1.0	1.9	3.1	19.5	
WSgust	525600.0	2.809195	0.0	1.3	2.6	4.1	26.6	
WSstdev	525600.0	0.47339	0.0	0.4	0.5	0.6	4.2	
WD	525600.0	153.435172	0.0	59.0	181.0	235.1	360.0	1
WDstdev	525600.0	8.582407	0.0	3.7	8.6	12.3	99.4	
BP	525600.0	994.197199	985.0	993.0	994.0	996.0	1003.0	
Cleaning	525600.0	0.000923	0.0	0.0	0.0	0.0	1.0	
Precipitation	525600.0	0.001905	0.0	0.0	0.0	0.0	2.5	
TModA	525600.0	35.246026	9.0	24.2	30.0	46.9	81.0	
TModB	525600.0	32.471736	8.1	23.6	28.9	41.5	72.5	
Comments	0.0	NaN	NaN	NaN	NaN	NaN	NaN	

Summary statistic for Togo Solar Dataset

In [21]: `togo.describe().T`

Out[21]:

	count	mean	min	25%	50%	75%	max	
Timestamp	525600	2022-04-25 12:00:30.000000768	2021-10-25 00:01:00	2022-01-24 06:00:45	2022-04-25 12:00:30	2022-07-25 18:00:15	2022-10-25 00:00:00	
GHI	525600.0	230.55504	-12.7	-2.2	2.1	442.4	1424.0	3
DNI	525600.0	151.258469	0.0	0.0	0.0	246.4	1004.5	2
DHI	525600.0	116.444352	0.0	0.0	2.5	215.7	805.7	1
ModA	525600.0	226.144375	0.0	0.0	4.4	422.525	1380.0	3
ModB	525600.0	219.568588	0.0	0.0	4.3	411.0	1367.0	
Tamb	525600.0	27.751788	14.9	24.2	27.2	31.1	41.4	
RH	525600.0	55.01316	3.3	26.5	59.3	80.8	99.8	
WS	525600.0	2.368093	0.0	1.4	2.2	3.2	16.1	
WSgust	525600.0	3.22949	0.0	1.9	2.9	4.4	23.1	
WSstdev	525600.0	0.55774	0.0	0.4	0.5	0.7	4.7	
WD	525600.0	161.741845	0.0	74.8	199.1	233.5	360.0	
WDstdev	525600.0	10.559568	0.0	6.9	10.8	14.1	86.9	
BP	525600.0	975.915242	968.0	975.0	976.0	977.0	983.0	
Cleaning	525600.0	0.000535	0.0	0.0	0.0	0.0	1.0	
Precipitation	525600.0	0.001382	0.0	0.0	0.0	0.0	2.3	
TModA	525600.0	32.444403	13.1	23.9	28.4	40.6	70.4	
TModB	525600.0	33.54333	13.1	23.6	28.4	43.0	94.6	
Comments	0.0	NaN	NaN	NaN	NaN	NaN	NaN	

Count missing values per column

In [25]: sierraleone.isna().sum()


```
Out[25]: Timestamp      0
        GHI             0
        DNI             0
        DHI             0
        ModA            0
        ModB            0
        Tamb            0
        RH              0
        WS              0
        WSgust          0
        WSstdev         0
        WD              0
        WDstdev         0
        BP              0
        Cleaning        0
        Precipitation    0
        TModA           0
        TModB           0
        Comments        525600
        dtype: int64
```

```
In [26]: benin.isna().sum();togo.isna().sum()
```

```
Out[26]: Timestamp      0
        GHI             0
        DNI             0
        DHI             0
        ModA            0
        ModB            0
        Tamb            0
        RH              0
        WS              0
        WSgust          0
        WSstdev         0
        WD              0
        WDstdev         0
        BP              0
        Cleaning        0
        Precipitation    0
        TModA           0
        TModB           0
        Comments        525600
        dtype: int64
```

```
In [27]: benin.isna().sum();togo.isna().sum()
```

```
Out[27]: Timestamp      0
        GHI             0
        DNI             0
        DHI             0
        ModA            0
        ModB            0
        Tamb            0
        RH              0
        WS              0
        WSgust          0
        WSstdev         0
        WD              0
        WDstdev         0
        BP              0
        Cleaning        0
        Precipitation    0
        TModA           0
        TModB           0
        Comments        525600
        dtype: int64
```

There are no missing values of in this dataset but if there were we can use this code snipt to check percentage of the missing values depends on our objective

```
missing_cols = df.columns[df.isna().mean() > 0.05] # Identify columns
with >5% nulls
print("Columns with >5% missing values:", missing_cols)`
```

Outlier Detection & Cleaning

```
In [28]: outlier_cols = ["GHI", "DNI", "DHI", "ModA", "ModB", "WS", "WSgust", "BP", "Precipita

sierraleone[outlier_cols] = sierraleone[outlier_cols].apply(zscore)
benin[outlier_cols]       = benin[outlier_cols].apply(zscore)
togo[outlier_cols]        = togo[outlier_cols].apply(zscore)
outliers_sierraleone = sierraleone[(sierraleone[outlier_cols].abs() > 3).any(axis=1)]
outliers_benin = benin[(benin[outlier_cols].abs() > 3).any(axis=1)]
outliers_togo = togo[(togo[outlier_cols].abs() > 3).any(axis=1)]
print("Outliers detected for Sierraleone solar dataset:", outliers_sierraleone.shape[0])
print("=====")
print("Outliers detected for Benin solar dataset:", outliers_benin.shape[0])
print("=====")
print("Outliers detected for Togo solar dataset:", outliers_togo.shape[0])
```

Outliers detected for Sierraleone solar dataset: 20747

=====

Outliers detected for Benin solar dataset: 9481

=====

Outliers detected for Togo solar dataset: 13707

Handling missing values

```
In [29]: sierraleone.fillna(sierraleone.median(), inplace=True) # Impute missing values with
benin.fillna(benin.median(), inplace=True)
togo.fillna(togo.median(), inplace=True)
```

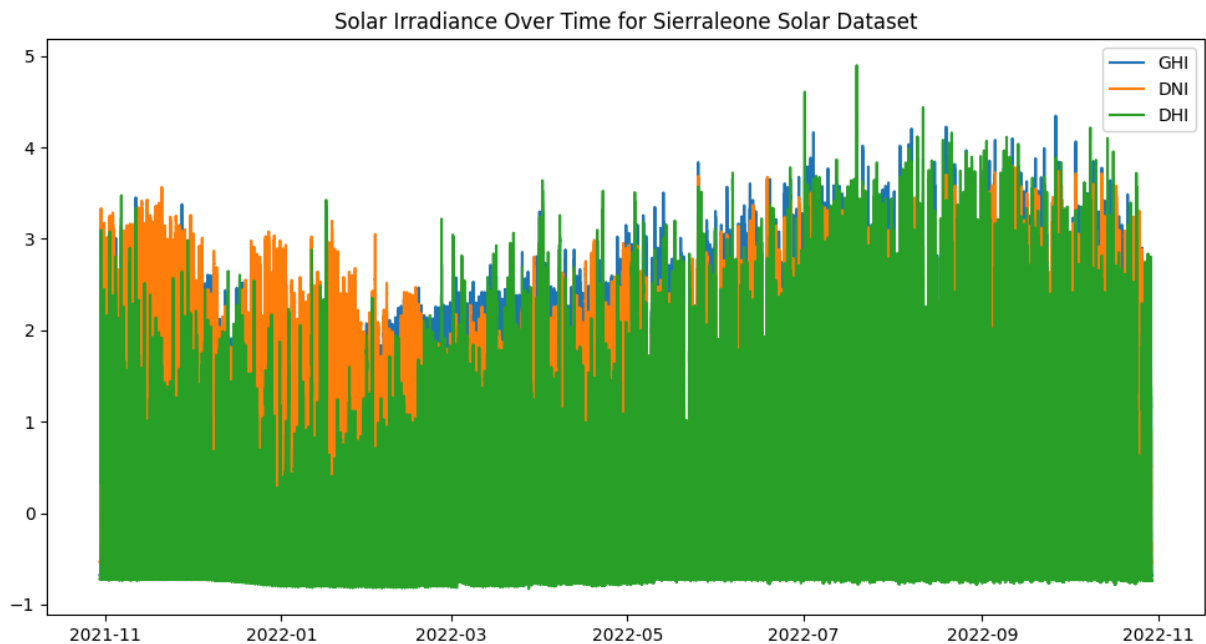
Export cleaned data

```
df.to_csv("C:\\Users\\girma\\OneDrive\\Desktop\\10 Academy\\week-1\\solar-
challenge-week-1\\data\\df_clean.csv", index=False)``
```

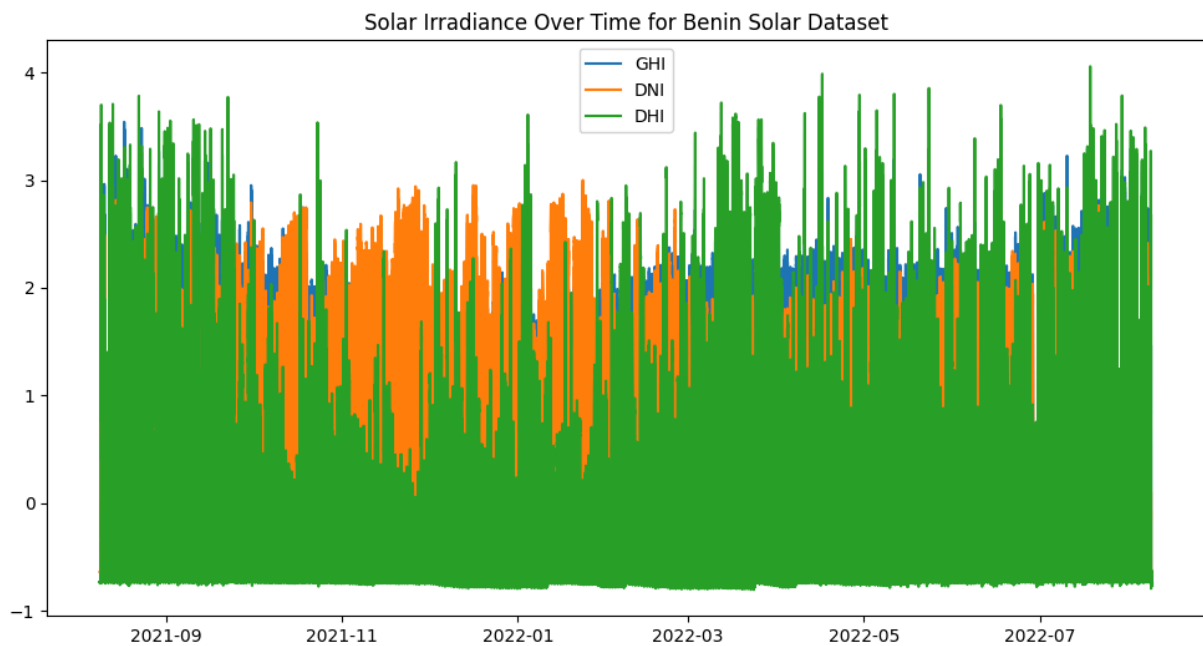
Time Series Analysis

Visualize trends over time:

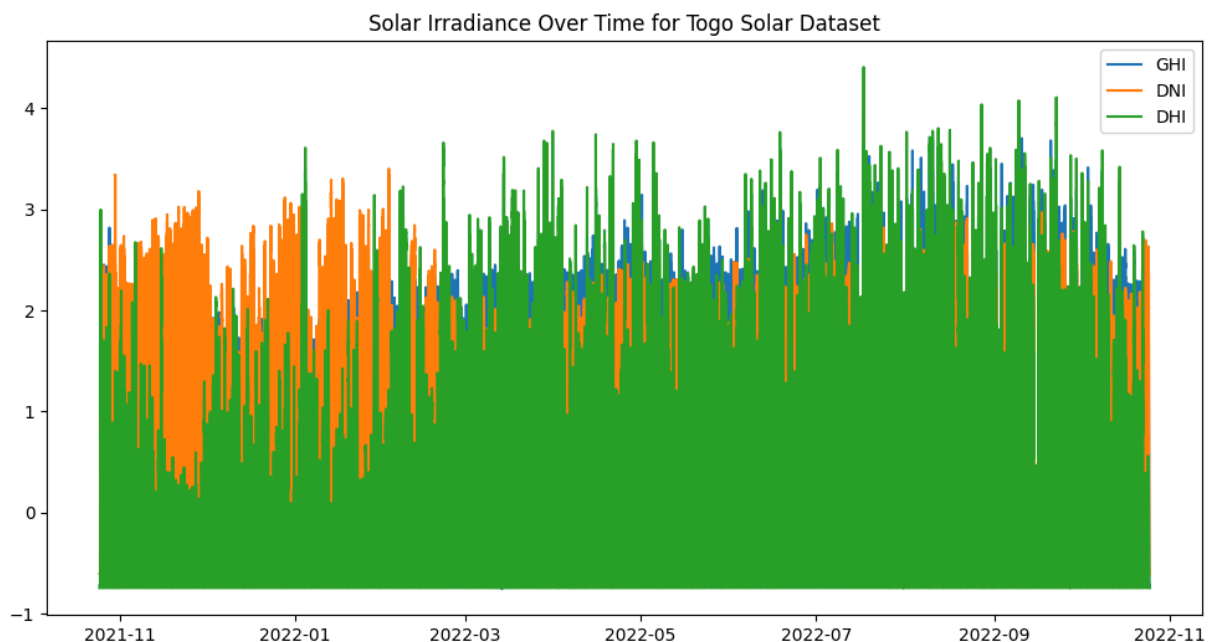
```
In [32]: sierraleone["Timestamp"] = pd.to_datetime(sierraleone["Timestamp"])
plt.figure(figsize=(12, 6))
plt.plot(sierraleone["Timestamp"], sierraleone["GHI"], label="GHI")
plt.plot(sierraleone["Timestamp"], sierraleone["DNI"], label="DNI")
plt.plot(sierraleone["Timestamp"], sierraleone["DHI"], label="DHI")
plt.legend()
plt.title("Solar Irradiance Over Time for Sierraleone Solar Dataset")
plt.show()
```



```
In [33]: benin["Timestamp"] = pd.to_datetime(benin["Timestamp"])
plt.figure(figsize=(12, 6))
plt.plot(benin["Timestamp"], benin["GHI"], label="GHI")
plt.plot(benin["Timestamp"], benin["DNI"], label="DNI")
plt.plot(benin["Timestamp"], benin["DHI"], label="DHI")
plt.legend()
plt.title("Solar Irradiance Over Time for Benin Solar Dataset")
plt.show()
```



```
In [34]: togo["Timestamp"] = pd.to_datetime(togo["Timestamp"])
plt.figure(figsize=(12, 6))
plt.plot(togo["Timestamp"], togo["GHI"], label="GHI")
plt.plot(togo["Timestamp"], togo["DNI"], label="DNI")
plt.plot(togo["Timestamp"], togo["DHI"], label="DHI")
plt.legend()
plt.title("Solar Irradiance Over Time for Togo Solar Dataset")
plt.show()
```

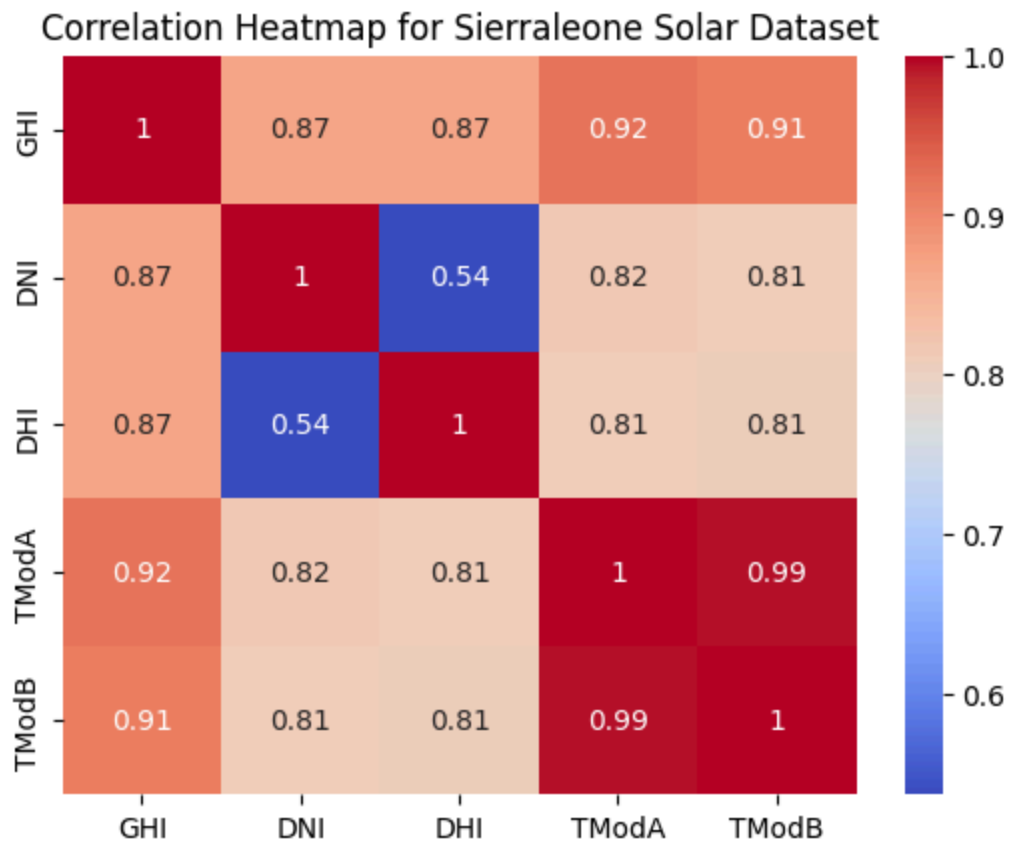


```
In [ ]:
```

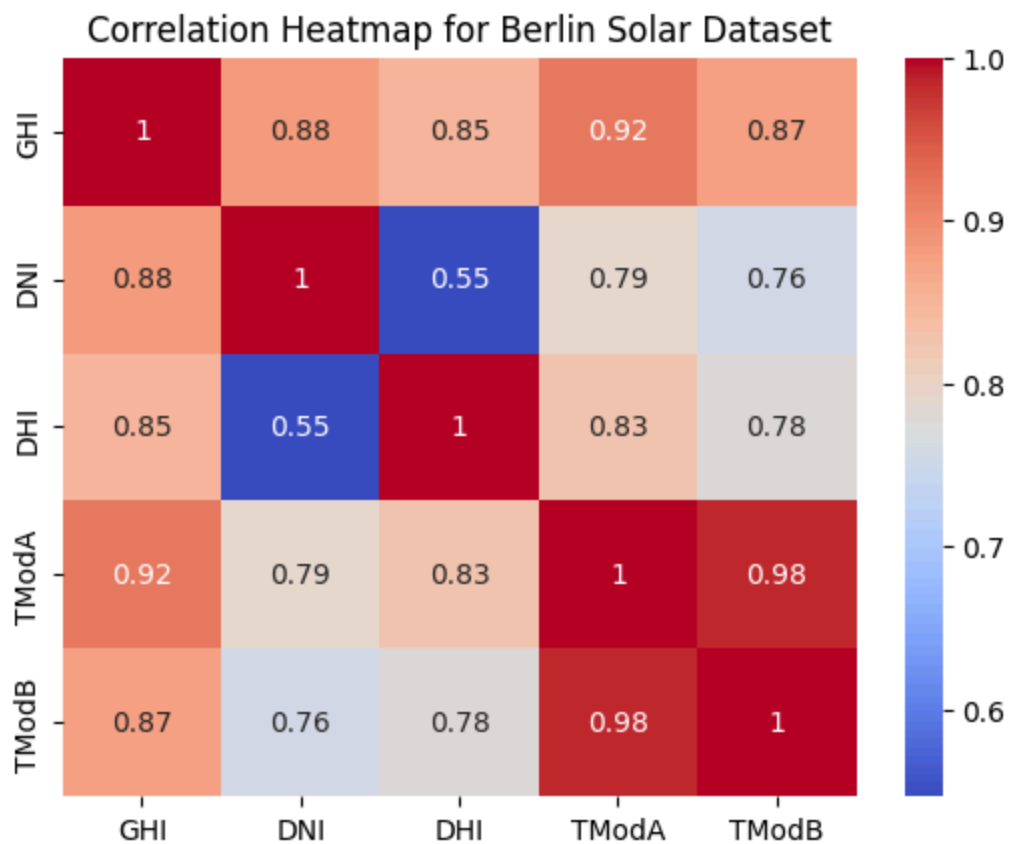
Correlation & Relationship Analysis

Generate a heatmap to visualize correlations:

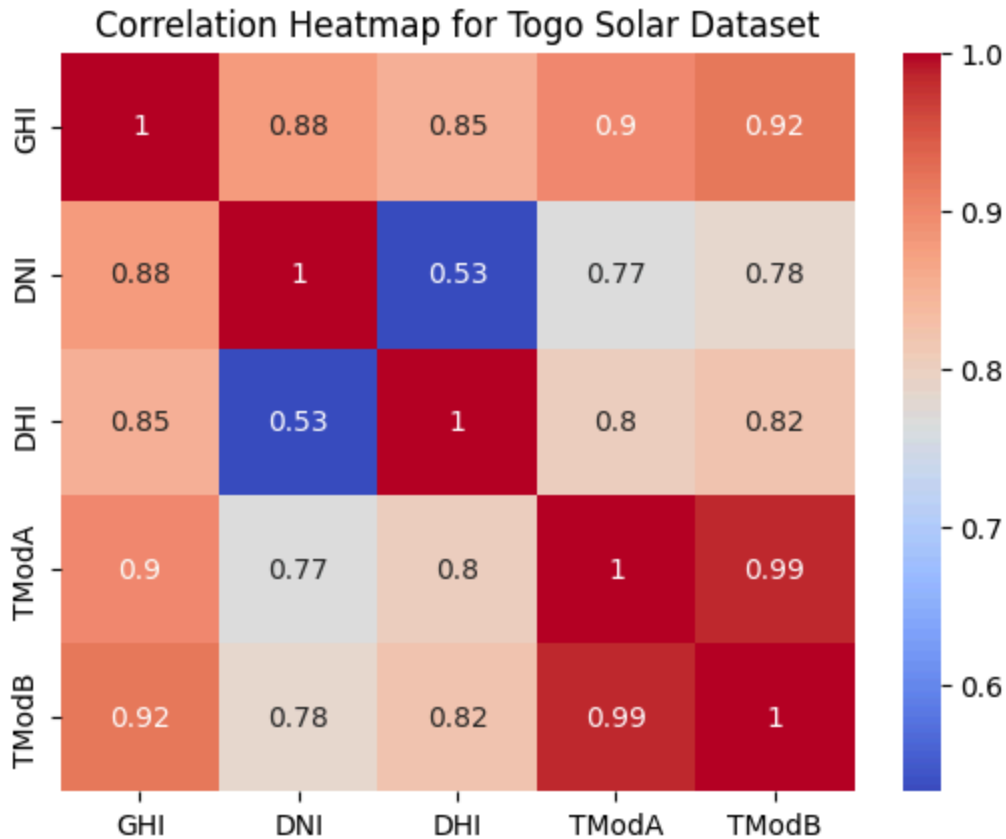
```
In [35]: corr_matrix_s = sierraleone[["GHI", "DNI", "DHI", "TModA", "TModB"]].corr()
sns.heatmap(corr_matrix_s, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap for Sierraleone Solar Dataset")
plt.show()
```



```
In [37]: corr_matrix_b = benin[["GHI", "DNI", "DHI", "TModA", "TModB"]].corr()
sns.heatmap(corr_matrix_b, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap for Berlin Solar Dataset")
plt.show()
```



```
In [38]: corr_matrix_t = togo[["GHI", "DNI", "DHI", "TModA", "TModB"]].corr()
sns.heatmap(corr_matrix_t, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap for Togo Solar Dataset")
plt.show()
```



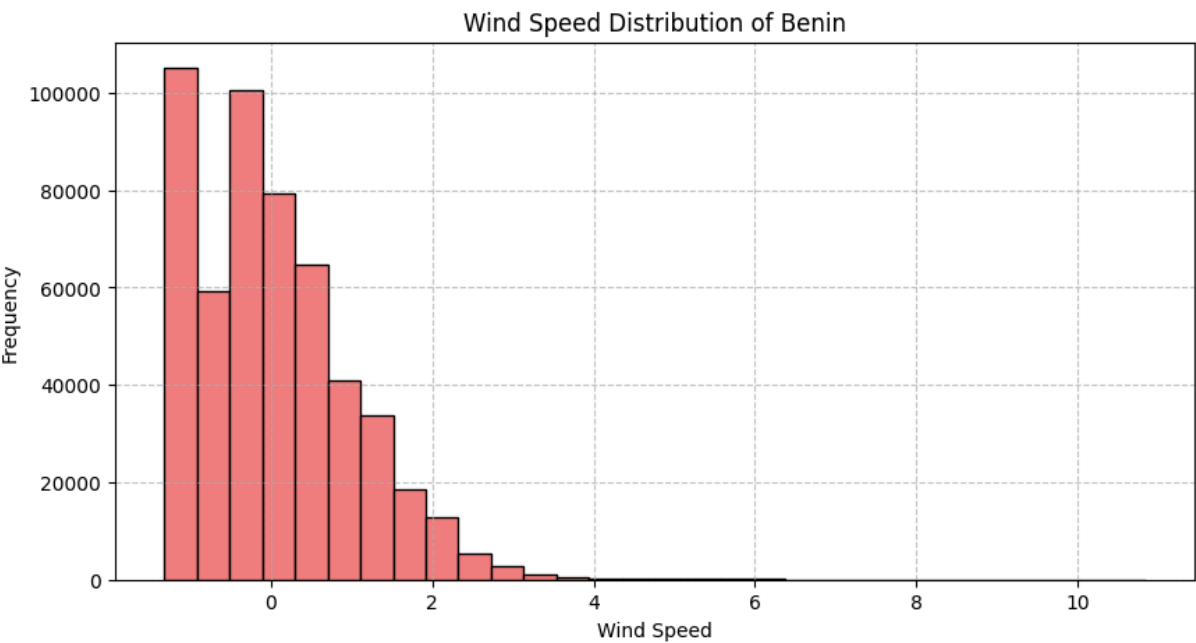
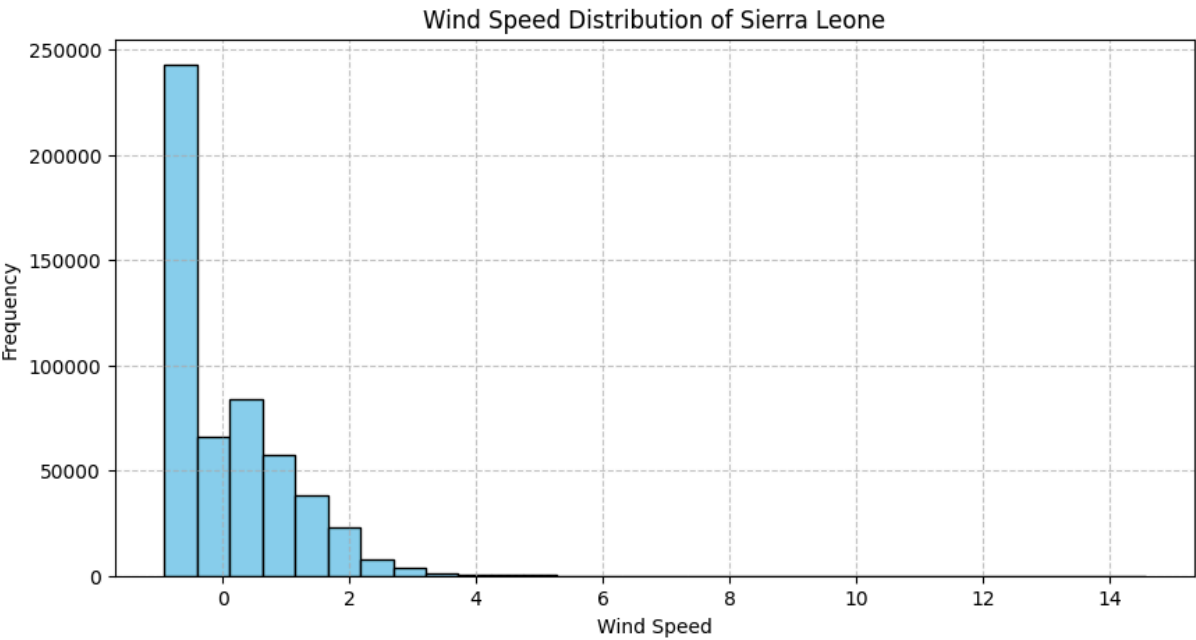
Wind & Distribution Analysis

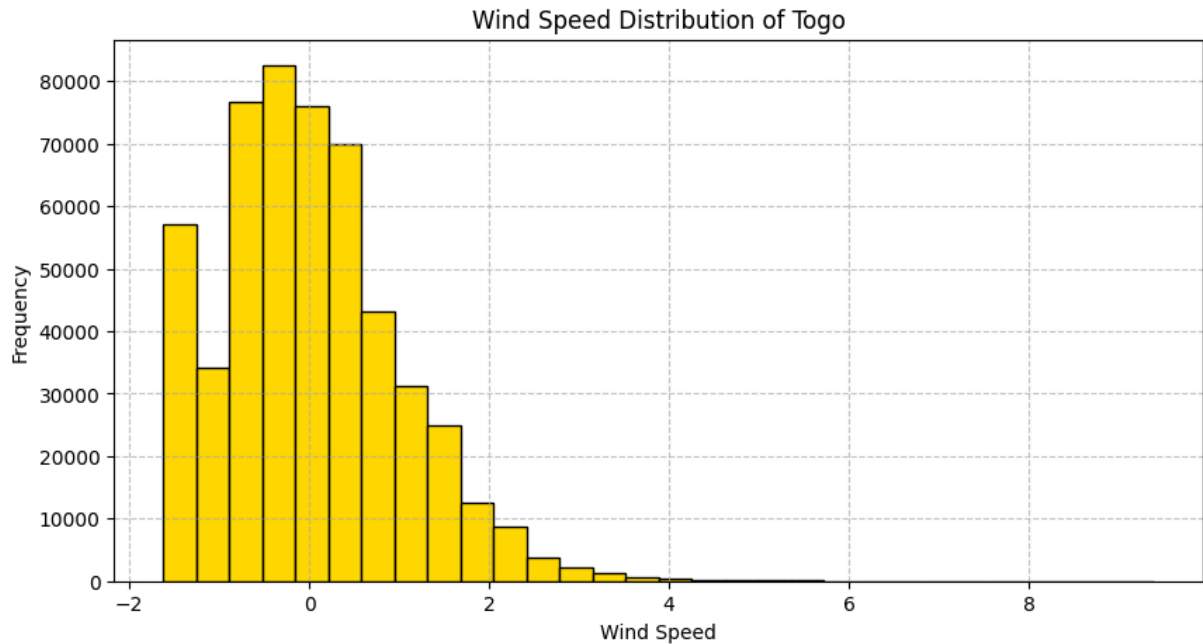
```
In [39]: # Sierra Leone
plt.figure(figsize=(10, 5))
sierraleone["WS"].hist(bins=30, color="skyblue", edgecolor="black")
plt.title("Wind Speed Distribution of Sierra Leone")
plt.xlabel("Wind Speed")
plt.ylabel("Frequency")
plt.grid(True, linestyle="--", alpha=0.7)
plt.show()

# Benin
plt.figure(figsize=(10, 5))
benin["WS"].hist(bins=30, color="lightcoral", edgecolor="black")
plt.title("Wind Speed Distribution of Benin")
plt.xlabel("Wind Speed")
plt.ylabel("Frequency")
plt.grid(True, linestyle="--", alpha=0.7)
plt.show()

# Togo
plt.figure(figsize=(10, 5))
togo["WS"].hist(bins=30, color="gold", edgecolor="black")
plt.title("Wind Speed Distribution of Togo")
plt.xlabel("Wind Speed")
plt.ylabel("Frequency")
```

```
plt.grid(True, linestyle="--", alpha=0.7)
plt.show()
```





Bubble Chart

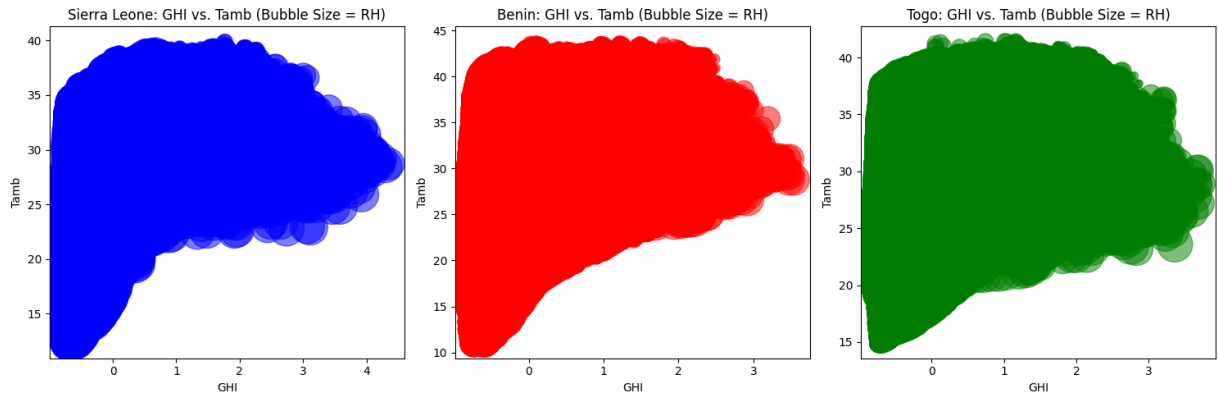
```
In [43]: # Create figure with 3 subplots
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(15, 5))

# Sierra Leone Bubble Chart
axes[0].scatter(sierraleone["GHI"], sierraleone["Tamb"], s=sierraleone["RH"]*10, alpha=0.5, color="red")
axes[0].set_xlabel("GHI")
axes[0].set_ylabel("Tamb")
axes[0].set_title("Sierra Leone: GHI vs. Tamb (Bubble Size = RH)")

# Benin Bubble Chart
axes[1].scatter(benin["GHI"], benin["Tamb"], s=benin["RH"]*10, alpha=0.5, color="red")
axes[1].set_xlabel("GHI")
axes[1].set_ylabel("Tamb")
axes[1].set_title("Benin: GHI vs. Tamb (Bubble Size = RH)")

# Togo Bubble Chart
axes[2].scatter(togo["GHI"], togo["Tamb"], s=togo["RH"]*10, alpha=0.5, color="green")
axes[2].set_xlabel("GHI")
axes[2].set_ylabel("Tamb")
axes[2].set_title("Togo: GHI vs. Tamb (Bubble Size = RH)")

# Adjust layout for clarity
plt.tight_layout()
plt.show()
```



Cross-Country Comparison

```
In [41]: # Create a figure with 3 subplots
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(15, 5))

# Plot DNI comparison
axes[0].boxplot([benin["DNI"], sierraleone["DNI"], togo["DNI"]], labels=["Benin", "Sierra Leone", "Togo"])
axes[0].set_title("DNI Comparison Across Countries")
axes[0].set_ylabel("DNI Values")

# Plot DHI comparison
axes[1].boxplot([benin["DHI"], sierraleone["DHI"], togo["DHI"]], labels=["Benin", "Sierra Leone", "Togo"])
axes[1].set_title("DHI Comparison Across Countries")
axes[1].set_ylabel("DHI Values")

# Plot GHI comparison
axes[2].boxplot([benin["GHI"], sierraleone["GHI"], togo["GHI"]], labels=["Benin", "Sierra Leone", "Togo"])
axes[2].set_title("GHI Comparison Across Countries")
axes[2].set_ylabel("GHI Values")

# Adjust layout for readability
plt.tight_layout()
plt.show()
```

C:\Users\girma\AppData\Local\Temp\ipykernel_1480\4271794736.py:5: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be dropped in 3.11.

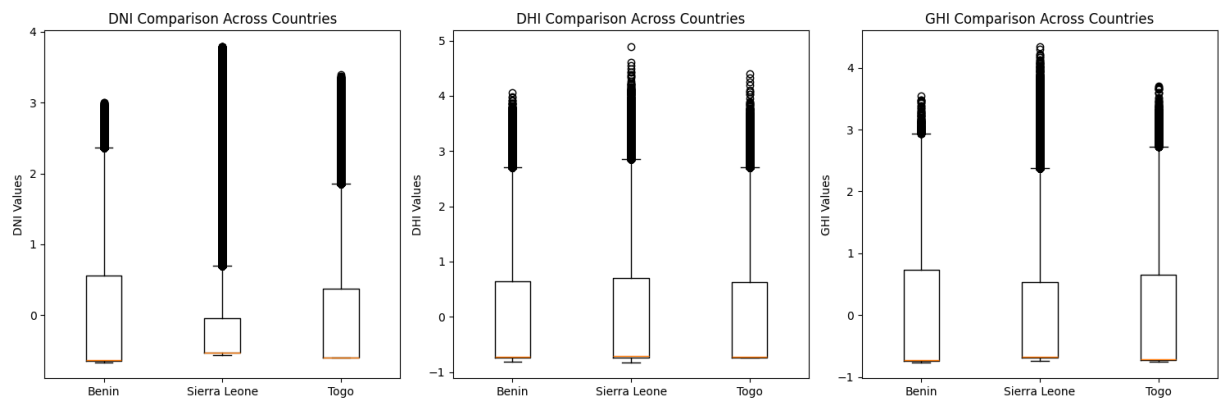
```
axes[0].boxplot([benin["DNI"], sierraleone["DNI"], togo["DNI"]], labels=["Benin", "Sierra Leone", "Togo"])
```

C:\Users\girma\AppData\Local\Temp\ipykernel_1480\4271794736.py:10: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be dropped in 3.11.

```
axes[1].boxplot([benin["DHI"], sierraleone["DHI"], togo["DHI"]], labels=["Benin", "Sierra Leone", "Togo"])
```

C:\Users\girma\AppData\Local\Temp\ipykernel_1480\4271794736.py:15: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be dropped in 3.11.

```
axes[2].boxplot([benin["GHI"], sierraleone["GHI"], togo["GHI"]], labels=["Benin", "Sierra Leone", "Togo"])
```



In []: